FEATURED PROJECT

Automatic Street-Lighting Control Syste

B.Sc. Electrical & Electronics Engineering -- AIUB

LDR-based	Automatic	Dimming logic
ambient-light sensing	dusk/dawn switching	cuts energy waste

THE PROBLEM

Streetlights typically run at full brightness all night, wasting significant energy during hours with little or no traffic -- a costly problem for municipalities and the grid.

HOW IT WORKS

- Ambient-light sensing: an LDR-based front end detects dusk and dawn, driving fully automatic ON/OFF switching with no manual intervention.
- Dimming logic: during low-traffic night hours the controller reduces lamp brightness, cutting power draw while maintaining safe visibility.
- Energy optimization: combined switching + dimming targets the two biggest sources of waste in conventional street lighting.

DESIGN HIGHLIGHTS

- Analog sensing chain: LDR signal conditioning with thresholding for reliable dusk/dawn detection, immune to transient light changes.
- Control logic: state-based switching and dimming behaviour designed for unattended outdoor operation.
- Practical constraints: designed around component cost, reliability, and ease of deployment on existing fixtures.

SKILLS DEMONSTRATED

Circuit Design | Analog Sensing | Control Logic | Energy Efficiency | Embedded Systems | Prototyping & Testing

WHY IT MATTERS

Some of the highest-impact engineering needs no cloud platform -- just careful analog and control design solving a real, everyday problem. Energy-efficient embedded design is a theme I want to keep building my career on.